

NYC Yellow Taxi

Ingestion, ETL & Analytics on HDFS + Spark

COURSE & CLASS

Course — Big Data Analytics

Class — BSCS-13A

Course Group — Group 3 (BDA, BSCS-13A)

Project Group # — 17

Institution — NUST SEECS

GROUP 17 · CONTRIBUTIONS

Malik Muhammad Hassaan — CMS 460233 · Data ingestion, validation, profiling

Muhammad Ahmed — CMS 456679 · ETL pipeline, business-question SQL, optimizations

Dataset Overview & Rationale

- **Source** — NYC TLC Yellow Taxi Trip Records
- **File** — yellow_tripdata_2024-01.parquet
- **Period** — January 2024 (one full month)
- **Volume** — 2,964,624 raw trip records
- **Schema** — 19 columns: datetime, numeric, categorical, location IDs
- **Format** — Parquet (binary columnar, compressed)
- **Provider** — nyc.gov / TLC Trip Record Data

WHY THIS DATASET

- Volume > 500K rows justifies HDFS + Spark
- Each row is a trip — a natural fact table
- Clear dimensions: time, location, payment, vendor
- Contains realistic data quality issues for ETL
- Public, well-documented, columnar format

Data Quality Issues & Cleaning

Identified during profiling (profile_data.py); resolved during ETL (etl.py).

ISSUE → ACTION

- **Negative fare amounts** — impute with distance-bin median fare
- **Zero passenger count** — set minimum to 1 passenger
- **Non-integer passenger count** — round to nearest integer
- **Duplicate trip rows** — dropDuplicates (keep first)
- **Trip distance outliers** — cap at 99th percentile

ISSUE → ACTION

- **Tip greater than fare** — cap tip at cleaned fare amount
- **Missing numeric values** — median imputation per column
- **Missing categorical codes** — fallback to 0 / mode
- **Datetime inconsistencies** — standardize to timestamp; derive duration
- **Future pickup timestamps** — replace with hour/weekday median

Pipeline Architecture

Ingest

ingest.py

Download · validate ·
upload to HDFS

Profile

profile_data.py

Schema · missing · stats ·
QA

ETL

etl.py (Spark)

Clean · transform · model
· partition

Analytics

analytics.py (Spark SQL)

5 business questions ·
charts

Report

build_final_report.py

Bundle CSVs + charts →
PDF

Storage layer: HDFS /warehouse/raw/... → /warehouse/processed/...

Compute layer: Apache Spark 3.5.8 (PySpark, Spark SQL, broadcast joins, partitioning, caching)

Dimensional Model (Star Schema)

dim_datetime

749 rows

dim_payment

5 rows

fact_trips

2,964,624 rows

trip distance · fare · tip · revenue · IDs

dim_trip_category

3 rows

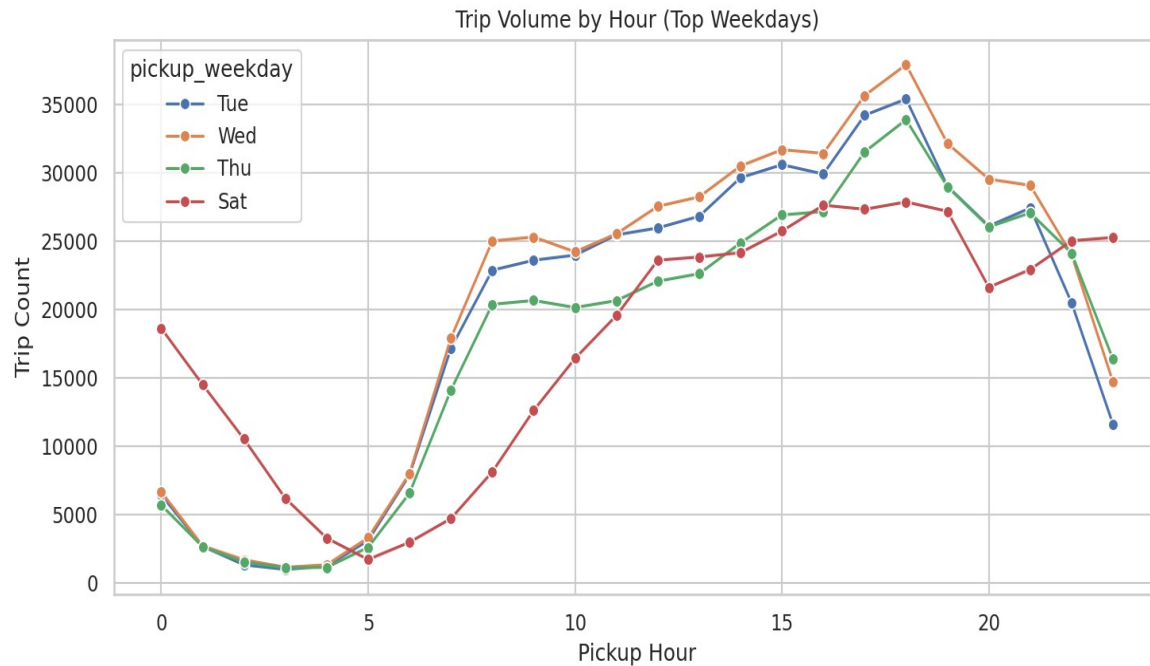
fact_trips joins to each dimension via natural keys (datetime_key, payment_type, distance_bucket).

Data Analytics & Visualization — 5 Business Questions

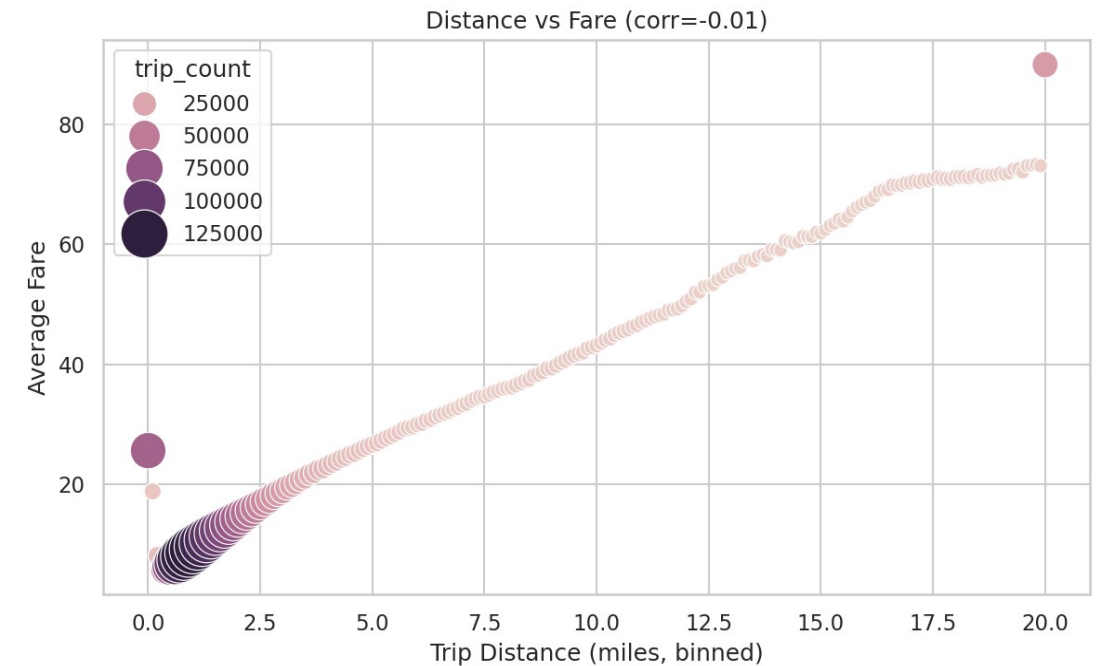
- Q1. How do trip volume and average fare vary by hour and day of week?
- Q2. What is the correlation between trip distance and fare amount?
- Q3. How does payment type preference change by time and location?
- Q4. How does average tip differ by payment type, distance, or vendor?
- Q5. Which pickup/dropoff location pairs have highest volume and fare?

Source: Dataset_Justification.pdf · Implemented as Spark SQL queries in analytics.py.

Q1 — Hourly Demand • Q2 — Distance vs Fare



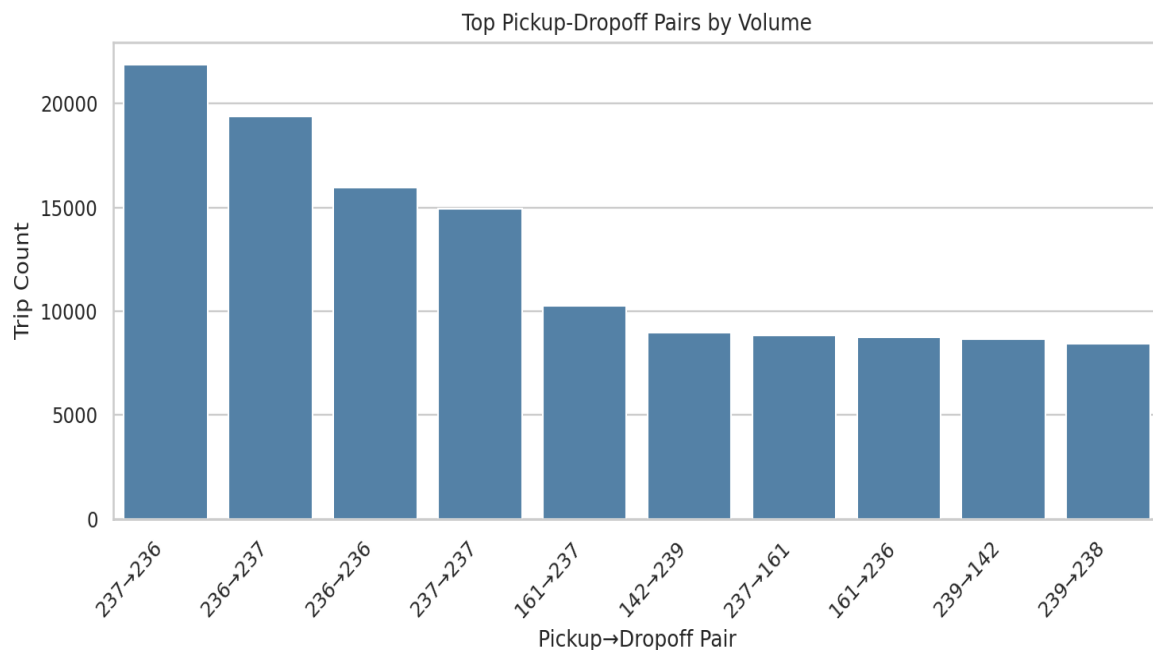
- Demand peaks 5–7 PM on weekdays.
- Fri/Sat sustain volume late at night.
- Lowest volume in the 3–5 AM window.



- Strong positive distance-vs-fare relationship.
- Within-bin correlation low (per-bin variance).
- Confirms distance as primary fare driver.

FINDINGS

Q3 — Payment · Q4 — Tipping · Q5 — Top Pairs



KEY NUMBERS

- **Q3 Payment** — Credit card $\approx$ 80–84% in core Manhattan zones
- **Q4 Tip (CC long)** — \$11.58 avg · 21.2% of fare
- **Q4 Tip (CC short)** — \$2.71 avg · 27.9% of fare
- **Q4 Tip (Cash)** — $\approx$ 0 (No-charge / Dispute $\rightarrow$ $\sim$ \$0.05)
- **Q5 Top pair** — 237 $\rightarrow$ 236 · 21,883 trips
- **Q5 Top revenue pair** — 132 $\rightarrow$ 132 · \$246,050 (JFK)

- Credit card dominates payment mix at high-volume zones during peak hours.
- Tipping rate (% of fare) is highest on short trips; tip dollar amount highest on long trips.
- A handful of pickup/dropoff pairs (Manhattan core + JFK) concentrate disproportionate revenue.

Pipeline Optimizations

- **Broadcast joins** — Small dim tables broadcast to executors -eliminates shuffle on fact ↔ dim joins.
- **Caching reused DataFrame** — Cleaned DF cached before deriving fact + dimensions to avoid recomputation.
- **Partitioned Parquet writes** — fact_trips & dim_datetime partitioned by pickup_year, pickup_month.
- **percentile_approx** — Fast quantile estimation for distance capping & median imputation.
- **Dynamic partition overwrite** — spark.sql.sources.partitionOverwriteMode=dynamic for incremental loads.
- **Tuned shuffle partitions** — spark.sql.shuffle.partitions=16 for the single-month workload.

Business Insights & Conclusion

- **Demand is highly time-bound** — Peak 5–7 PM and pre-dawn lulls suggest dynamic supply / surge windows.
- **Distance is the dominant fare driver** — Strong positive trend supports predictable, distance-based pricing.
- **NYC taxis are effectively cashless** — Credit card holds 80%+ share in the busiest zones — invest in card UX.
- **Tip behaviour is payment-driven** — Card riders tip materially; cash / no-charge segments tip near zero.
- **Revenue concentrates in a handful of OD pairs** — Manhattan core (161/236/237) and JFK (132) drive disproportionate revenue.
- **Data quality discipline pays off** — ETL preserved all 2.96M trips while neutralizing nulls, outliers and impossible values.

Dataset & GitHub Repository

DATASET

NYC TLC Yellow Taxi Trip Records (Jan 2024)

<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Direct file: yellow_tripdata_2024-01.parquet

GITHUB REPOSITORY

NYC-Taxi-Ingestion-Pipeline

<https://github.com/hassaanawan48/NYC-Taxi-Ingestion-Pipeline>

Includes ingest.py, profile_data.py, etl.py, analytics.py, build_final_report.py and PDFs.

Thank you. • Questions are welcome.